PKCERT AI & Software Development Internship

Task 12 – Clustering & Dimensionality Reduction

Objective:
Develop practical skills in unsupervised machine learning using clustering algorithms and dimensionality reduction techniques.

Total Marks: 100

Part A – Dataset Selection & Preparation (15 Marks)

1. Select a public dataset suitable for clustering.

2. Describe the dataset and preprocess it, including handling missing values and feature scaling.

Part B – Clustering Techniques (45 Marks)

1. Implement K-Means clustering and determine an appropriate number of clusters.

2. Implement Hierarchical Clustering and visualize the dendrogram.

3. Compare the clustering results and discuss observations.

Part C – Dimensionality Reduction (30 Marks)

1. Apply PCA to reduce dimensions and visualize the transformed data.

2. Apply t-SNE for two-dimensional visualization.

3. Compare PCA and t-SNE, highlighting advantages, limitations, and use cases.

Part D – Analysis & Conclusion (10 Marks)

1. Interpret the results and recommend the most suitable techniques for the selected dataset with justification.

Assessment Area	Marks
Dataset Preparation	15
Clustering Techniques	45
Dimensionality Reduction	30
Analysis & Conclusion	10
Total	100